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EXTENDED ABSTRACT

Evaluation of Deep Learning Models for Satellite IR Precipitation Estimation

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1. Introduction and objectives

Accurate precipitation estimation is critical for climate research, hydrology, and disaster management, especially in regions with sparse in situ observations. Satellite-based observations, particularly infrared (IR) brightness temperatures, offer frequent, wide-area coverage and near-real-time monitoring capabilities. IR sensors detect thermal emissions primarily from cloud tops, which correlate indirectly with surface rainfall. However, this indirect relationship poses significant challenges for quantitative precipitation estimation (QPE), as cloud-top temperatures may not always reflect the intensity or location of rainfall at the surface. Factors such as cloud type, microphysical processes, and atmospheric humidity further complicate the mapping from IR signals to precipitation rates.

Comparisons across studies are challenging due to heterogeneity in regions, precipitation regimes, reference datasets, evaluation metrics, and IR sensors, preventing robust conclusions. To address this, we conduct a systematic inter-comparison of state-of-the-art deep learning models using a common dataset, reference targets, and metrics. The objective is to identify modeling strategies for IR-based QPE, taking into account the intermittence of rain and extreme events, requiring tailored loss functions and data augmentation.

2. Methods and data

We use seven infrared (IR) channels (6.2–13.4 μm) from the SEVIRI instrument on the Meteosat Second Generation (MSG) satellite. These channels capture thermal emissions primarily from cloud tops and water vapor content across different atmospheric layers, providing indirect but valuable information for surface precipitation estimation.

To estimate precipitation from IR-only data, we implement and compare several deep learning approaches. Deterministic models, including U-Net [1] and DeepLabV3+ [2], are designed to optimize pixel-wise accuracy and minimize pointwise errors. In contrast, generative models, such as conditional GANs [3] and diffusion models [4], focus on capturing realistic spatial patterns including extreme rainfall events, aiming to reproduce the heavy-tailed distributions.

Training and validation use a 13-year radar mosaic from Météo-France co-located with the IR data, covering a wide range of meteorological conditions. The long temporal coverage allows robust evaluation of model performance for both typical and extreme precipitation events.

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3. Preliminary results and/or expected outcomes

Initial experiments show that deterministic models produce accurate mean precipitation estimates but underestimate extreme events. Generative models better capture heavy rainfall and realistic spatial patterns, though pixel-wise errors are higher, with lower CSI and F1 scores and higher RMSE and MAE. These challenges reflect both the weak correlation between IR brightness temperatures and surface rainfall and the heavy-tailed nature of extreme precipitation events.

4. Discussion

These preliminary results highlight the potential of deep learning to improve IR-only precipitation retrievals, particularly in capturing variability and extreme events. Future work will extend models to the full MSG swath, incorporate temporal dynamics, and explore hybrid approaches combining deterministic and generative strengths.

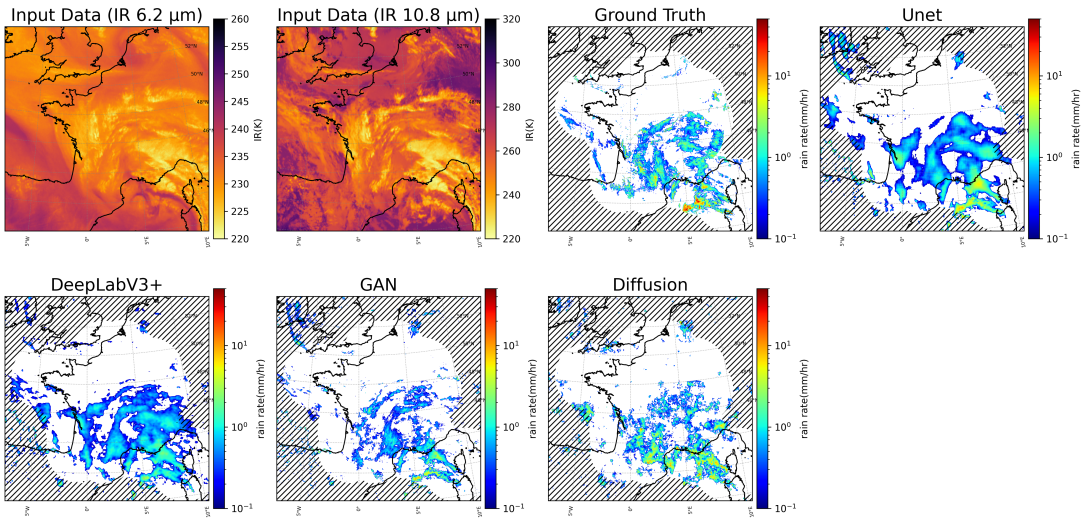

Figure 1. *Qualitative comparison of precipitation estimates from deterministic (U-Net, DeepLabV3+) and generative (cGAN, diffusion models)..*

References

[1] Wang, D., Yang, S., Li, X., Peng, J., Ma, H., & Wu, X. Multi-scale Attention-UNet-based near-real-time precipitation estimation from FY-4A/AGRI and Doppler radar observations. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 1–14 (2024). https://doi.org/10.1109/JSTARS.2024.3488854

[2] Berthomier, L., & Perier, L. Espresso: A global deep learning model to estimate precipitation from satellite observations. *Meteorology* **2**(4), 421–444 (2023). https://www.mdpi.com/2674-0494/2/4/25

[3] Han, K.-H., Jeong, J., & Hong, S. IMERG-like precipitation retrieval from Geo-Kompsat-2A observations using conditional generative adversarial networks. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* **18**, 14467–14479 (2025). https://doi.org/10.1109/JSTARS.2025.3575763

[4] Ho, J., Jain, A., & Abbeel, P. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems*, 33, 6840–6851 (2020).